

Question: 1

Use the ping command to check the connectivity between your computer and www.spotify.com. Note the response times and explain what they indicate about your network connection.

Answer: 1

Spotify Connectivity Test Log (Windows Command Prompt)

To test the connectivity to [Spotify](https://www.spotify.com), open the Command Prompt on your computer and execute the core network diagnostic query:

```
cmd
```

```
ping www.spotify.com
```

Expected Diagnostic Output:

```
text
```

```
Pinging www.spotify.com with 32 bytes of data:
```

```
Reply from 104.199.65.124: bytes=32 time=42ms TTL=57
```

```
Reply from 104.199.65.124: bytes=32 time=40ms TTL=57
```

```
Reply from 104.199.65.124: bytes=32 time=45ms TTL=57
```

```
Reply from 104.199.65.124: bytes=32 time=41ms TTL=57
```

```
Ping statistics for 104.199.65.124:
```

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
```

```
Approximate round trip times in milli-seconds:
```

```
    Minimum = 40ms, Maximum = 45ms, Average = 42ms
```

What the Response Times Indicate

- **Excellent Network Health:** The **0% packet loss** indicator confirms a highly reliable, error-free physical connection path between your home network adapter and Spotify's Content Delivery Network (CDN) edge servers.
- **Low Latency (42ms Average):** A round-trip time (RTT) sitting between **30ms to 60ms** is standard and highly efficient for geographical routing from India to regional servers. This low latency indicates that the fiber-optic infrastructure, local ISPs, and wide-area routing links are processing data without congestive bottlenecks.
- **Stable Connection (Low Jitter):** The minimal variance between the minimum latency (**40ms**) and maximum latency (**45ms**) indicates a highly stable internet stream. This

steady performance ensures smooth, uninterrupted music or audio playback on your device, free of loading stalls or buffering lag.

Question: 2

Your friend is unable to access www.zomato.com, but other websites work fine. Use nslookup to check the DNS resolution for www.zomato.com and describe what the output tells you.

Hint: Look for the IP address and any error messages in the nslookup result.

Answer: 2

Zomato DNS Resolution Test Log (Windows Command Prompt)

To diagnose why your friend cannot access [Zomato](http://www.zomato.com) while other websites load fine, open the Command Prompt and run the standard DNS lookup diagnostic tool:

```
cmd
```

```
nslookup www.zomato.com
```

Expected Successful Diagnostic Output:

```
text
```

```
Server: UnKnown
```

```
Address: 192.168.1.1
```

```
Non-authoritative answer:
```

```
Name: www.zomato.com
```

```
Addresses: 104.18.23.118
```

```
104.18.22.118
```

```
Use code with caution.
```

What the Output Tells You About the Problem

If your console outputs a valid non-authoritative answer showing real public IP addresses (like the Cloudflare IP blocks 104.18.22.118 or 104.18.23.118), it tells you that **the DNS resolution is working perfectly**.

Since the internet can successfully translate the domain name into a machine-readable IP address, your friend's problem is **not a DNS failure**. Instead, the issue points to one of the following root causes:

- **Local Browser Cache Issue:** Your friend's browser might have cached a corrupted web file or cookie from a previous session. Clearing the browser cache or trying to load Zomato in an Incognito window will usually fix this.
 - **IP-Level Routing Block:** Your friend's specific IP address might have been temporarily flagged or blocked by Zomato's security firewall (such as Cloudflare protection) due to unusual network behavior on their internet connection.
 - **Local Firewall / Antivirus Restriction:** A security software suite or router configuration on your friend's local computer might be accidentally blocking traffic targeting that specific domain range.
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Question: 3

Open Wireshark and capture network traffic while you refresh your Instagram feed in a browser. Identify and list the different protocols you see in the capture.

Hint: Filter by protocol type (like HTTP, TCP, DNS) in Wireshark to make it easier.

Answer: 3

Captured Network Protocols for Instagram Traffic

When you open Wireshark and capture traffic while refreshing your feed on [Instagram](#) inside a web browser, the network stream relies on a suite of modern web protocols.

By using the search filter bar at the top of Wireshark to group the captured traffic, you will identify and log the following protocols:

- **DNS (Domain Name System):**
 - **What you see:** Before the webpage loads, you will spot UDP packets querying domain nodes like `://instagram.com` or `://instagram.com`.
 - **Role:** Translates the human-readable Instagram domain into a machine-routable public IP address.
- **TCP (Transmission Control Protocol):**
 - **What you see:** A high volume of packets logging the initial **[SYN, ACK]** connection handshakes and subsequent segment exchanges.

- **Role:** Acts as the underlying transport mechanism, guaranteeing reliable, error-free delivery of large app image assets and web structures.
 - **TLS / HTTPS (Transport Layer Security):**
 - **What you see:** Heavy chunks of traffic labeled "**TLSv1.3**" or "**Encrypted Application Data**" inside the protocol tracking column.
 - **Role:** Protects your login session tokens, secure cookies, and interactive feed inputs using deep encryption so that no external packet sniffer can read your personal account updates.
 - **QUIC / HTTP/3:**
 - **What you see:** If using a modern web browser like Google Chrome or Microsoft Edge, a large portion of the stream will show up as **UDP-based QUIC traffic**.
 - **Role:** Used heavily by Instagram's parent company (Meta) to stream multimedia components like Reels, profile photos, and stories with minimal loading delay or network lag.
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Question: 4

A user reports that their Flipkart app is slow to load product images. Use netstat to list all current network connections on your system and identify if there are any suspicious or high-usage connections that could be affecting bandwidth.

Answer: 4

Steps to Run Netstat and Check Active Connections

To identify if any hidden network connections or background processes are consuming your internet bandwidth and slowing down the **Flipkart app**, follow these diagnostic steps:

- **Step 1: Open Command Prompt as Administrator:**
Press the Windows Key, type **cmd**, right-click on **Command Prompt**, and select **Run as administrator**.
- **Step 2: Execute the Active Connection Command:**
Type the command below into the terminal window and press Enter:

```
cmd
```

```
netstat -ano
```

Sample Netstat Diagnostic Log Output

The terminal output will list active TCP and UDP streams on your computer network interface cards:

```
text
```

Active Connections

Proto	Local Address	Foreign Address	State	PID
TCP	192.168.1.15:53124	104.16.249.249:443	ESTABLISHED	4124
TCP	192.168.1.15:53128	142.250.190.46:443	ESTABLISHED	2840
TCP	192.168.1.15:53140	198.51.100.72:443	ESTABLISHED	8816
UDP	192.168.1.15:61005	*:*		2840

How to Identify Bandwidth-Hogging Connections

To find out what is causing the slowdown based on the log entries above, analyze the connections using these criteria:

- **Legitimate Application Traffic (PID 4124 & 2840):**
The connections terminating at port **:443** (HTTPS) for PIDs like 4124 and 2840 are standard web browser traffic streams (e.g., pulling product assets or Google Search requests). They indicate standard network activity.
 - **Pinpointing the Suspect Connection (PID 8816):**
The connection running to the foreign address **198.51.100.72** under **PID 8816** stands out. If you notice a single unknown foreign IP exchanging high volumes of data, it indicates a background update or hidden application consuming your bandwidth.
 - **Finding and Stopping the Process:**
To see exactly what application owns that network stream, open Windows **Task Manager** (Ctrl + Shift + Esc), click on the **Details** tab, and sort the list by the **PID** column. Find **8816** to identify the application. If it is an unneeded app (like an automated cloud storage sync or a hidden torrent program running in the background), right-click it and select **End Task** to instantly free up your bandwidth for the Flipkart app.
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Question: 5

Your college WiFi keeps disconnecting during IPL match streaming. Create a troubleshooting checklist using ping, tracert, and netstat to systematically diagnose the issue step by step. For each tool, write what command you would use and what result would indicate a problem.

Answer: 5

IPL Streaming Wi-Fi Troubleshooting Checklist

During high-traffic events like live IPL match streaming, college Wi-Fi networks frequently suffer from bandwidth exhaustion, wireless signal interference, or router overload. Follow this step-by-step checklist to systematically diagnose where the connection is breaking:

Step 1: Verify Local Wi-Fi Gateway Stability (Using Ping)

- **Command to Use:**

cmd

ping -t 192.168.1.1

(Replace 192.168.1.1 with your specific college Wi-Fi router's local gateway IP address. The -t flag runs a continuous test).

- **What Indicates a Problem:**
 - **High Latency or Jitter:** Response times jumping wildly from **5ms to over 500ms** mean the wireless access point is overloaded with too many students streaming the match at once.
 - **"Request timed out" or Packet Loss:** Regular connection drops mean your phone or laptop is losing its physical wireless signal due to heavy radio interference in the room.
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Step 2: Locate Where the Internet Stream Slows Down (Using Tracert)

- **Command to Use:**

cmd

tracert ://jiocinema.com

(This tracks every network hop from your device to the streaming server).

- **What Indicates a Problem:**
 - **High Delays on Internal Hops (Hops 2–4):** If latency spikes immediately within the first few hops inside the college network, the main campus core switch or network firewall is struggling to handle the traffic.
 - **Star Icons (* * *) near the Destination:** If the traceroute fails completely outside the college network, it indicates a wide-area network (WAN) routing problem or an outage with the college's main Internet Service Provider (ISP).
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Step 3: Check for Background Bandwidth Hogs (Using Netstat)

- **Command to Use:**

cmd

netstat -ano

(Lists all active network connections alongside their Process IDs).

- **What Indicates a Problem:**
 - **Too Many Active Connections:** Seeing dozens of unexpected connections in the **ESTABLISHED** state means background apps (like cloud syncing tools, system updates, or peer-to-peer apps) are quietly running on your device.
 - **Bandwidth Starvation:** These hidden background connections eat up your limited network allocation slot, starving your streaming player of the bandwidth it needs and causing the stream to freeze or buffer.
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